Zihong Luo

Incoming M.S. Student in AI Engineering – ECE, Carnegie Mellon University

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Education

Carnegie Mellon University *Pittsburgh, PA, USA*

M.S. in Artificial Intelligence Engineering – Electrical and Computer Engineering *Aug 2026 – Dec 2027 expected*

University of Liverpool *Liverpool, UK*

BSc in Computing Science | ***GPA: 3.87/4.0*** | *First-Class Honours* *Sep 2023 – Jun 2026*

Research Experience

SmartLab, University of Liverpool *Liverpool, UK*

Research Assistant, Advisor: Prof. Anh Nguyen *Oct 2025 – May 2026*

- Worked on robot learning for manipulation, focusing on gripper-aware vision-language-action policy modeling and complex robotic grasping.
- Contributed to **GVLA**, a gripper-aware VLA framework that conditions manipulation policies on gripper morphology, visual observations, and language instructions.
- Studied how end-effector structure, scene objects, and task affordances can be represented for more generalizable robotic manipulation.
- Supported benchmark construction for complex grasping, covering object detection, semantic reasoning, grasp planning, and trajectory-level execution.
- Research outcomes include one **ECCV 2026** paper on gripper-aware VLA models and one **IROS 2026** paper on complex robotic grasping.

Jifu Medical, AI Algorithm Group *Shenzhen, China*

Algorithm Intern – Robotics & Multimodal Perception *Jun 2025 – Aug 2025*

- Built a robot imitation-learning data pipeline based on **ACT**, **ALOHA**, and **LeRobot** for SO-101 dual-arm manipulation.
- Implemented teleoperation, synchronized multi-view recording, and action-state logging for real-robot demonstration collection.
- Collected and validated 50+ manipulation episodes, preparing datasets for behavior cloning and vision-language-action policy learning.

MBZUAI *Abu Dhabi, UAE*

Research Assistant, Advisor: Prof. Imran Razzak *Dec 2024 – Jul 2025*

- Developed **PG-SAM**, a prior-guided medical image segmentation framework using medical large language models to provide modality-aware semantic priors.
- Designed a Modality Prior Aligner and fusion decoder to connect high-level semantic reasoning with pixel-level dense prediction.
- Outcome: paper accepted to **IEEE BIBM 2025**.

University of Exeter *Exeter, UK*

Summer Research Assistant, Advisor: Prof. Yanda Meng *Mar 2024 – Aug 2024*

- Contributed to **IMDR**, a multimodal representation learning framework for ophthalmic disease grading under incomplete and noisy modality settings.
- Designed ablation studies and incomplete-modality evaluations to analyze modality-shared and modality-specific representations.
- Outcome: paper accepted as an **AAAI 2025 Oral**.

Selected Publications

* Equal contribution. Author order follows the original paper.

Gripper-aware Vision-Language-Action Models [ECCV] *2026*

– Hanyi Zhang*, **Zihong Luo**, et al.

Beyond Visual Grasping: Benchmarking Complex Grasping from Detection to Execution [IROS] *2026*

– Hanyi Zhang*, ..., **Zihong Luo**, et al.

PG-SAM: Prior-Guided SAM with Medical LLMs for Multi-organ Segmentation [BIBM] *2025*

– Yiheng Zhong*, **Zihong Luo***, et al.

Incomplete Modality Disentangled Representation for Ophthalmic Disease Diagnosis [AAAI Oral] *2025*

– Chengzhi Liu*, Zile Huang*, ..., **Zihong Luo**, et al.

MC-DBN: Modality Completion with Deep Belief Networks [ICPR] *2024*

– **Zihong Luo***, Chengzhi Liu*, et al.

Additional Peer-Reviewed Publications *2024 – 2026*

– Co-authored additional papers in multimodal learning, medical AI, and molecular representation learning, including AI4Physics ICML Workshop 2026 and BMC Medical Informatics 2024.

Robotics Platforms & Projects

Quadruped Locomotion Robustness | *Unitree Go2, MuJoCo, PPO, ONNX* *2026*

– Evaluated robust quadruped locomotion policies under transient external force perturbations, focusing on recovery behavior, compliance boundaries, and tail-risk safety.

– Developed disturbance-probing protocols to analyze whether residual policy adaptation can improve force-safety under 30N external perturbations.

– Investigated the gap between generic disturbance recovery and strict transient force-compliance for real-world legged robot deployment.

Real-Robot Manipulation with Multi-Gripper Platforms | *xArm7, Franka Panda, UR5, RGB-D* *2026*

– Worked with multi-gripper real-robot platforms, including parallel-jaw, suction, soft-fin, and dexterous grippers.

– Used dual-camera RGB-D observations to support visual grounding, demonstration collection, and gripper-aware policy evaluation.

Robot Learning and Simulation Workflows | *MuJoCo, Isaac Lab, Isaac Sim, LeRobot* *2025*

– Built workflows for teleoperation, data collection, behavior cloning, reinforcement learning, and sim-to-real prototyping.

– Collected and processed 50+ SO-101 manipulation episodes for imitation-learning experiments.

Self-Balancing Wheeled-Legged Robot | *C++, Arduino, ESP32, Control* *2025*

– Built a wheeled-legged robot with inverse kinematics, IMU feedback, and cascaded PID control for stable remote-controlled locomotion.

Skills

Programming: Python, C++, MATLAB, Java, Git, Linux
Machine Learning: PyTorch, TensorFlow, OpenCV, Transformers, Imitation Learning, Reinforcement Learning
Robotics: ROS/ROS2, MuJoCo, Isaac Sim, Isaac Lab, LeRobot, URDF, IK/FK, PID Control
Hardware: Arduino, ESP32, Raspberry Pi, RGB-D Cameras, IMU, LiDAR